EngageNet

M2 Presentation - April 26, 2026

Lado Turmanidze, Luka Javakhishvili, Mariami Gadelia, Keso Chikhladze

Supervisors: Dr. Philipp Müller & Beso Mikaberidze

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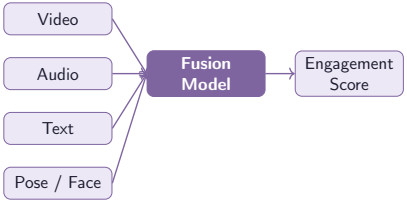
Today's Agenda

1. **Refresher** - what we are doing and the BiMamba-TTA base architecture
2. **Regression adaptation** - why Gaussian fails and our Beta distribution framework
3. **TTA for regression** - Beta-based uncertainty, sample selection, and the test-time loss
4. **Architectural contribution** - fixing the order-sensitivity of inter-modal fusion via Gumbel-Sinkhorn

Part 1: Refresher - What We Are Doing

Predicting Engagement in Expert-Novice Interactions

- Inputs: video, audio, text, body pose, facial landmarks - all fused together.
- **Task:** predict a continuous score $y \in [0, 1]$ frame-wise at **25 Hz**.
- Two corpora: **NoXi** (Europe, FR/EN/DE) and **NoXi+J** (Japan/China).
- Key challenge: **cross-cultural domain shift** - train on NoXi, test on NoXi+J.



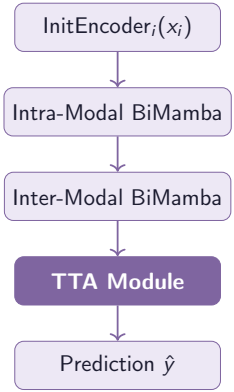
Refresher - BiMamba-TTA Architecture

Based on *BiMamba-TTA* (Jia et al., NeurIPS 2025) - originally designed for emotion *classification*.

Two modules:

1. **Intra-Modal BiMamba** - bidirectional SSM; captures temporal context.
2. **Inter-Modal BiMamba** - sweep *across* modalities to share cross-stream information.

A **TTA module** then adapts the model at test time using only the test samples themselves.



Refresher - Intra-Modal BiMamba

For each modality i, the encoder produces $h_i \in \mathbb{R}^{L' \times C'_i}$. The BiMamba block then:

(1) Gating

$g_i = \sigma(W_i^g h_i + b_i^g) \quad \text{(SiLU)}$
Highlights engagement-relevant features.

(2) Bidirectional SSM scan

$h_i^{\rightarrow} = g_i \otimes \mathrm{SSM}_{\rightarrow}(\cdots)$
$h_i^{\leftarrow} = g_i \otimes \mathrm{SSM}_{\leftarrow}(\mathrm{Rev}_t(\cdots))$
$\otimes$ = Hadamard product.

(3) Residual output

$$u_i = h_i + \left(W_i^o \frac{h_i^{\rightarrow} + \mathrm{Rev}_t(h_i^{\leftarrow})}{2} + b_i^o \right)$$

Refresher - Inter-Modal BiMamba & TTA

Inter-Modal BiMamba

Concatenate outputs and transpose:

$$m = (u_1 \circ \cdots \circ u_M)^\top \in \mathbb{R}^{(\sum C_i') \times L'}$$

A second BiMamba sweeps across modalities.

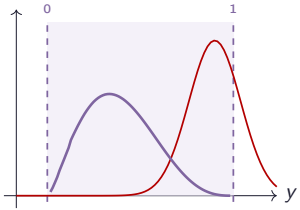
Test-Time Adaptation

Smart filter: select samples that are confident overall but uncertain unimodally. **Surgical fine-tuning:** only update BatchNorm and the first layers. Core weights stay **frozen**.

Everything above was originally designed for *classification*. We adapt it to *regression*.

Part 2: Why Gaussian Fails for Engagement

$$p(y \mid x) = \mathcal{N}(\hat{\mu}(x), \hat{\sigma}^2(x))$$



The problem

Engagement is **strictly bounded**: $y \in [0, 1]$.
A Gaussian assigns probability to $(-\infty, 0)$ and $(1, \infty)$ - **probability leakage**.
Poorly calibrated near boundaries.

The Beta Regression Framework

Support on $[0, 1]$:

$$p(y \mid \alpha, \beta) = \frac{y^{\alpha-1}(1-y)^{\beta-1}}{B(\alpha, \beta)}$$

Output shape parameters via softplus:

$$\alpha = \mathrm{sp}(z_\alpha) + 1, \quad \beta = \mathrm{sp}(z_\beta) + 1$$

Predictive mean:

$$\hat{\mu} = \frac{\alpha}{\alpha + \beta}$$

Boundary-aware uncertainty

Variance naturally vanishes at boundaries:

$$\lim_{\hat{\mu} \to 0} \mathrm{Var}[y] = 0$$

Training loss

Beta negative log-likelihood:

$$\mathcal{L}_{\mathrm{NLL}} = -(\hat{\alpha}-1)\ln y - (\hat{\beta}-1)\ln(1-y) + \ln B(\hat{\alpha}, \hat{\beta})$$

TTA - Sample Selection & Adaptive Thresholds

Two-level filter

$$S(x) = \mathbb{I}\{\, U_{\text{multi}}(x) \le \gamma_m(t) \;\; \text{AND} \;\; \sum w_i\, U_i(x) \ge \gamma_u(t) \,\}$$

Uncertainty decomposition (MC Dropout)

$$U_{\text{total}} = \underbrace{\frac{1}{T}\sum_t \mathrm{Var}[\text{Beta}_t]}_{\text{aleatoric}} + \underbrace{\frac{1}{T}\sum_t (\hat{\mu}^{(t)} - \bar{\mu})^2}_{\text{epistemic}}$$

We **adapt on high-epistemic samples**.

TTA - Test-Time Loss

Mutual Information Sharing (MIS) loss

$$\mathcal{L}_{\text{mis}} = \sum_{i=1}^{M} D_{KL}\big(\text{Beta}(\alpha_i, \beta_i) \parallel \text{Beta}(\alpha_{\text{multi}}, \beta_{\text{multi}})\big)$$

$$D_{KL}(B_1 \| B_2) = \ln \frac{B(\alpha_2, \beta_2)}{B(\alpha_1, \beta_1)} + (\alpha_1 - \alpha_2)\psi(\alpha_1) + (\beta_1 - \beta_2)\psi(\beta_1) + (\alpha_2 - \alpha_1 + \beta_2 - \beta_1)\psi(\alpha_1 + \beta_1)$$

Final TTA objective

$$\mathcal{L}_{\text{TTA}} = \mathcal{L}_{\text{mis}} + \lambda \sum_{i=1}^{M} \text{NLL}_{\text{Beta}}\big(\hat{\mu}_{\text{multi}},\ \alpha_i, \beta_i\big)$$

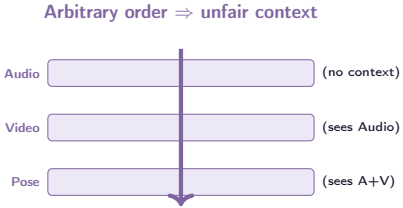
Part 4: The Order-Sensitivity Problem

The inter-modal BiMamba is biased by an *arbitrary* modality order

Proposition. For any SSM with $\bar{A} \neq I$, the output is **not permutation-invariant**.
Sketch: with $M = 2$,

$$y_2 - y_2' = C\bar{B}(\bar{A} - I)(u_1 - u_2) \neq 0$$

Vanishes only if $\bar{A} = I$ (memoryless map).



Fix: Differentiable Modality Ordering via Gumbel-Sinkhorn

(1) Score matrix. Affinities:

$$Z_{ij} = \frac{q_i^\top k_j}{\sqrt{d_s}}, \quad q_i = W_Q \bar{u}_i, \; k_j = W_K \bar{u}_j$$

(2) Gumbel perturbation.

$$\tilde{Z}_{ij} = \frac{Z_{ij} + G_{ij}}{\tau}$$

(3) Sinkhorn normalisation.

$$P = \mathrm{Sinkhorn}(\exp(\tilde{Z}))$$

(4) Soft reordering.

$$\tilde{u}_i = \sum_{j=1}^{M} P_{ij}\, u_j, \quad \tilde{m} = \mathrm{Trans}(\tilde{u}_1 \circ \cdots \circ \tilde{u}_M)$$

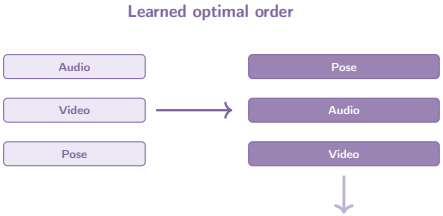
Equivariance Guarantee & Intuition

Equivariance guarantee

Relabelling modalities by $\rho \in S_M$ transforms the permutation as $\pi^*_\rho = \rho \circ \pi^*$. Representation H is **invariant to concatenation order**.

Connection to mHC

Uses the same Sinkhorn-Knopp machinery as our mHC residual idea.



Summary & Next Steps

What we formalised

Beta head,
variance-as-entropy proxy,
and surgical fine-tuning.

Novel architectural contribution

Permutation-equivariant inter-modal fusion via Gumbel-Sinkhorn.

Next steps

Evaluate NoXi $\rightarrow$ NoXi+J with/without TTA.

Time to implement!